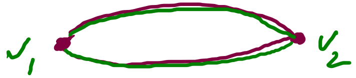
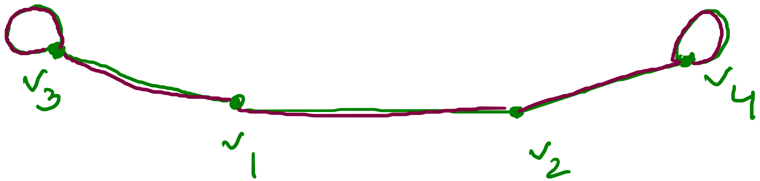
10.5 Euler + Hamiltonian paths.

(*) **Theorem 1** → A connected multigraph has an Euler path and no Euler circuit iff there are exactly 2 vertices of odd degree.

(*) **Theorem-2** → A connected multigraph with at least 2 vertices each having even degree has an Euler circuit.

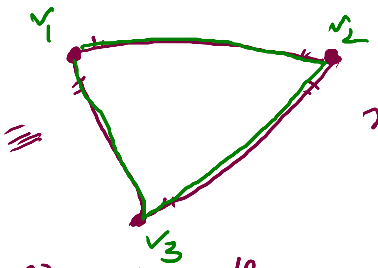


$v_1 v_2 v_1 \longrightarrow Th_2$



$v_3 v_3 v_1 v_2 v_4 v_4 \longrightarrow Th_1$

Dirac's theorem If G is a simple graph, with n-vertices and $n \geq 3$ such that the degree of every vertex is at least $n/2$ then G has a HC.

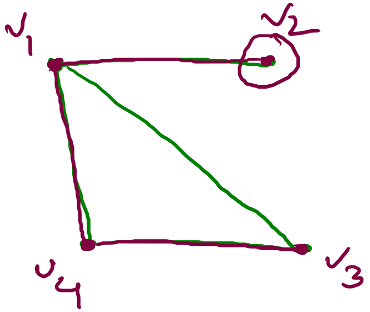


$n = 3$

$x \geq (n/2) = 1.5$

→ HC? Yes

→ $v_1 \; v_2 \; v_3 \; v_1$



$n = 4$

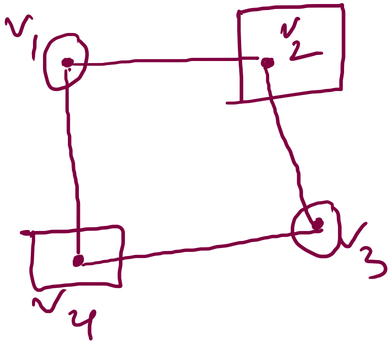
$x \geq n/2 = 2$

HC → No

HP → Yes

→ $v_2 v_1 v_3 v_4$

Ore's theorem if G is a simple graph, with n-vertices and $n \geq 3$ such that for every pair of non-adjacent vertices (u, v) $\deg(u) + \deg(v) \geq n$ then G has a HC.



$n = 4 > 3$

$$\left.\begin{array}{l} (v_1, v_3) = \deg(v_1) + \deg(v_3) = 2+2 = \underline{4} = n \\ (v_2, v_4) = \deg(v_2) + \deg(v_4) = 2+2 = \underline{4} = n \end{array}\right\} \text{Has HC}$$